TUTORIAL – 10: Automation & Regression Analysis

Aim

To analyze the suitability of software modules for **Automation Testing** and perform **Regression Testing** on a sample web application by identifying automation candidates, preparing a regression test suite, and evaluating automation feasibility.

Description

Automation Testing is a software testing technique in which test cases are executed automatically using testing tools instead of being executed manually. It is useful for repetitive, stable, and frequently executed test cases.

Regression Testing is performed after fixing bugs or modifying the software to ensure that existing functionalities continue to work correctly and that no new defects have been introduced.

Benefits of Automation Testing

- Saves testing time.
- Reduces human errors.
- Supports repeated execution.
- Improves test coverage.
- Generates faster reports.

Benefits of Regression Testing

- Verifies that recent code changes do not affect existing features.
- Ensures software stability after updates.
- Detects side effects of modifications.

Common Automation Testing Tools

- Selenium
- Cypress
- Playwright
- TestNG
- JUnit
- PyTest

Problem Statement
Analyze the modules of the **Student Management System** and identify which modules are suitable for automation testing. Execute a regression test suite after a software update and prepare an Automation Feasibility Report.

How to Perform This Practical (Student Procedure)
Step 1: Study the Application
Consider the **Student Management System** having the following modules:
- Login
- Registration
- Student Details
- Search Student
- Logout

Step 2: Analyze Each Module
Check:
- Is the module critical?
- Is it executed frequently?
- Does it change frequently?
- Is it suitable for automation?

Step 3: Identify Automation Candidates
Modules such as Login and Registration are executed repeatedly and are stable, making them good candidates for automation.

Step 4: Prepare Automation Candidate List
Create a table indicating:
- Test Case ID
- Functionality
- Automation Recommended
- Reason
- Automation Tool

Step 5: Execute Regression Test Cases
After fixing a bug or updating the application:

1. Execute all previously passed test cases.
2. Verify that existing features still work correctly.
3. Record the execution status.

Step 6: Prepare Regression Test Suite
Include:
- Login
- Registration
- Search
- Logout

Step 7: Prepare Automation Feasibility Report
Analyze:
- Repetitive test cases
- Stable functionality
- Business priority
- Frequency of execution
- Recommended automation tool

Implementation Details
The Student Management System is analyzed to identify modules suitable for automation testing. Regression testing is then performed to verify that recent code changes have not affected the existing functionalities.

Parameter Details

Parameter	Details
Project Name	Student Management System
Application/Software	Student Management System
Module Name	Login Module
SDLC Model	Waterfall Model
Programming Language	Python

IDE/Tool Used	Visual Studio Code / PyCharm
Input/Test Data	admin/admin123, admin/12345

Source Code

```
# Login Module

username = input("Enter Username: ")
password = input("Enter Password: ")

if username == "admin" and password == "admin123":
    print("Login Successful")
    print("Dashboard Opened")
else:
    print("Invalid Username or Password")
```

Module Analysis Table

Module Name	Functionality Critical (Yes/No)	Frequently Changed (Yes/No)	Suitable for Automation (Yes/No)	Reason
Login	Yes	No	Yes	Frequently executed and stable
Registration	Yes	No	Yes	Validation testing is repetitive
Student Details	Yes	Yes	No	Frequent business logic changes

Search Student	Yes	No	Yes	Repeated search operations
Logout	No	No	Yes	Simple repetitive functionality

Automation Candidate Identification

Test Case ID	Functionality	Automation Recommended (Yes/No)	Reason	Automation Tool
TC001	Login	Yes	Frequently executed	Selenium
TC002	Registration	Yes	Stable functionality	Selenium
TC003	Search Student	Yes	Repetitive operation	Selenium
TC004	Student Details Update	No	Business logic changes frequently	Manual Testing
TC005	Logout	Yes	Simple repetitive task	Selenium

Test Case Detailed

Field	Details
Test Case ID	TC001
Requirement ID	R001
Module Name	Login Module
Feature Under Test	User Authentication
Test Scenario	Login with valid username and password
Test Objective	Verify successful login after software update
Test Type (Unit/Integration/System/UAT)	Regression Testing
Test Technique (Black Box/White Box)	Black Box Testing
Priority	High
Severity	Critical
Preconditions	Application is deployed and Login page is available
Postconditions	User enters Dashboard
Test Environment	Windows 10
Browser/OS (if applicable)	Google Chrome
Test Data	admin / admin123
Expected Result	Login Successful
Actual Result	Login Successful

Status (Pass/Fail)	Pass
Executed By	Student Name
Execution Date	DD/MM/YYYY

Test Case Execution

Step No.	Test Steps	Input	Expected Result	Actual Result	Status
1	Launch Application	-	Login Page displayed	Login Page displayed	Pass
2	Enter Username	admin	Username accepted	Username accepted	Pass
3	Enter Password	admin123	Password accepted	Password accepted	Pass
4	Click Login	Click	Dashboard Opens	Dashboard Opens	Pass
5	Verify Existing Features	-	Features work correctly	Features working correctly	Pass

Regression Test Suite

Regression Test ID	Module	Scenario	Expected Result	Status
RT001	Login	Valid Login	Login Successful	Pass
RT002	Registration	New User Registration	Registration Successful	Pass
RT003	Search Student	Search Existing Student	Student Record Displayed	Pass

RT004	Logout	Logout	Return to Login Page	Pass

Automation Feasibility Report

Criteria	Observation
Repetitive Test Cases	Yes
Stable Functionality	Yes
High Business Priority	Yes
Frequently Executed	Yes
Suitable Automation Tool	Selenium
Recommendation	Automate Login, Registration, Search, and Logout modules

Regression Testing Summary

Metric	Value
Total Modules Analyzed	5
Automation Candidates	4
Regression Test Cases	4
Modules Excluded	1 (Student Details Update)
Recommendation	Automate stable and repetitive modules; continue manual testing for frequently changing modules

Observation

Sr. No.	Observation	Remarks
1	Login module suitable for automation	Yes
2	Registration module suitable for automation	Yes
3	Student Details Update requires manual testing	Business logic changes frequently
4	Regression test cases executed successfully	Pass
5	Existing functionalities unaffected after update	Completed

Output

Student Management System

Automation Analysis Completed

Automation Candidates

✔ Login
✔ Registration
✔ Search Student
✔ Logout

Regression Testing Completed

Total Regression Test Cases : 4
Passed : 4
Failed : 0

Overall Result

Automation Recommended

Regression Testing : PASS

Application Stable After Update

Solve the Below Mention Questionnaires

1. Which testing technique verifies functionality without examining source code?

A) White-Box Testing
B) Integration Testing
C) Black-Box Testing
D) Unit Testing
Answer: C) Black-Box Testing

2. Which testing technique requires knowledge of internal program code?

A) System Testing
B) Black-Box Testing
C) Acceptance Testing
D) White-Box Testing
Answer: D) White-Box Testing

3. Integration Testing is mainly used to verify:

A) Database Performance
B) Communication between modules
C) Source Code Formatting
D) User Interface Design
Answer: B) Communication between modules

4. System Testing validates:

A) Individual Functions Only
B) Single Module Performance

C) Complete Application Behavior
D) Database Tables Only
Answer: C) Complete Application Behavior

5. Which testing checks execution paths and branch coverage?
A) Black-Box Testing
B) White-Box Testing
C) Regression Testing
D) Usability Testing
Answer: B) White-Box Testing

Conclusion / Outcome

- Understood the concepts of **Automation Testing** and **Regression Testing**.
- Identified modules suitable for automation based on stability, frequency of execution, and business importance.
- Prepared an **Automation Feasibility Report** and a **Regression Test Suite**.
- Executed regression test cases to ensure that existing functionalities remained unaffected after software updates.
- Learned how automation testing improves testing efficiency while regression testing ensures application stability after changes.